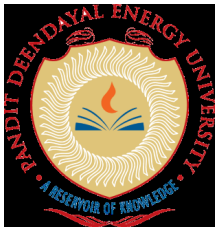


Department of Mechanical Engineering
School of Technology



Laboratory Report

Computational Fluid Dynamics and Heat Transfer Lab

Python and Google Colab Edition

Name of Course: _____

Course Code: _____

Semester: _____

Branch: _____

Batch (Division): _____

Name of Student: _____

Roll Number: _____

Academic Year: _____

Certificate

This is to certify that Mr./Miss. _____

Roll Number __ of B.Tech. Mechanical Engineering has satisfactorily completed the laboratory work in

20ME406P Computational Fluid Dynamics and Heat Transfer Lab

for the term ending in _____.

Date: _____

_____

Signature of Teacher

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Course Information

Pandit Deendayal Energy University
School of Technology, Department of Mechanical Engineering
20ME406P Computational Fluid Dynamics and Heat Transfer Lab

Teaching Scheme				Examination Scheme		
L	T	P	C	Lab Work	Lab Exam/Viva	Total
0	0	2	1	25	25	50

Course objectives

1. Use computational tools to study fluid-flow and heat-transfer problems.
2. Apply numerical discretization and iterative solution methods to governing equations.
3. Interpret convergence, conservation, stability, boundedness, and discretization error.
4. Post-process scalar and vector fields and compare numerical results with analytical, empirical, or benchmark data.

Course outcomes

CO1	Understand computational methods for fluid-flow analysis.
CO2	Apply computational methods to heat-transfer analysis.
CO3	Implement and compare numerical schemes for governing equations.
CO4	Select discretization and solution methods based on the problem physics.
CO5	Analyse and interpret numerical results using plots and field visualization.
CO6	Validate results against analytical solutions, correlations, or benchmark data.

Student Quick Start

Required route: Google Colab. No local software installation is required.

1. Open https://colab.research.google.com/ in a web browser and sign in using a Google account.
2. Select **File** > **Upload notebook** and upload the assigned notebook from the `colab_notebooks` folder.
3. Immediately select **File** > **Save a copy in Drive**. Work only on the saved copy.
4. Set `RUN_MODE = "quick"` for an initial execution check. Use `lab` or an approved `custom` case for the final report.
5. Select **Runtime** > **Run all**. The notebook reconstructs and runs the tested solver automatically.
6. Inspect the numerical summary, convergence plots, quantitative comparisons, and inline field contour.
7. In the final download cell, set `DOWNLOAD_RESULTS = True` and run the cell. Retain the downloaded ZIP before closing Colab.

No installation is required

- Do not install Python, Jupyter, NumPy, Matplotlib, a C++ compiler, or any VTK library.
- Use the default Colab CPU runtime. Selecting a GPU runtime does not accelerate these NumPy teaching solvers.
- Colab runtime storage is temporary. Unsaved files are deleted when the runtime disconnects or resets.

Submission for every experiment

- Saved Colab notebook with all required cells executed.
- Downloaded experiment results ZIP.
- Parameter set, grid, convergence/error plot, and quantitative result table.
- At least one correctly labelled inline field figure. ParaView figures are optional unless specifically assigned.
- Summary text file and answers to the compulsory assignment.
- A technically specific conclusion discussing accuracy, stability, conservation, and model limitations where relevant.

Model scope. Experiments 1, 3, 8, 10, and the natural-convection branch of Experiment 11 are deliberately reduced teaching models. They must not be described as industrial or high-fidelity CFD simulations.

1 Experiment 1: External Flow Analysis over a NACA 0012 Airfoil

Aim

To simulate incompressible external flow over a NACA 0012 airfoil and study the effect of angle of attack on velocity, pressure, wake structure, and qualitative force trends.

Governing Model

$$\nabla \cdot \mathbf{u} = 0, \tag{1}$$

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} = -\nabla p + \frac{1}{Re} \nabla^2 \mathbf{u}. \tag{2}$$

Boundary Conditions

Uniform inflow; convective/zero-gradient outflow with pressure reference; uniform far-field velocity on the upper and lower boundaries; no-slip enforced through the immersed mask.

Numerical Method

Cartesian immersed-geometry projection method. The NACA profile is represented by a smooth solid mask; penalization enforces near-zero velocity inside the body and a pressure-Poisson correction reduces divergence.

> **Model scope.** Teaching-level immersed-boundary solver. The force coefficients are qualitative and must not be presented as validated aerodynamic coefficients.

Run in Google Colab

Upload the notebook below to Google Colab, save a copy in Drive, select the run mode, and use **Runtime > Run all**.

```
colab_notebooks/01_NACA0012_Airfoil.ipynb
```

Argument order

```
nx ny Re alpha_deg iterations poisson_iterations
```

Primary outputs

```
airfoil_final.vtk, airfoil_wall.vtk, airfoil_mask.vtk, history.csv, airfoil_summary.txt
```

Procedure

1. Run the notebook first in `quick` mode to verify the Colab workflow.
2. Change to `lab` mode and select **Runtime > Run all**.
3. Confirm that the solver completed without exceptions and read the summary file.
4. Inspect the residual, error, conservation, or force history plotted in the notebook.
5. Use the inline VTK field preview to produce the required visualization. Downloaded VTK files may be opened in ParaView as an optional extension.
6. Modify only the parameter cell for the compulsory assignment; retain the original base-case results.

Observations to Record

Angle of attack, final continuity residual, maximum speed, wake pattern, pressure-side/suction-side behaviour, qualitative lift and drag trends.

Compulsory Assignment

Run $\alpha = -5°, 0°, 5°, 10°$ at $Re = 100$. Compare wake size, pressure distribution, final residual, and qualitative force trends.

Implementation Files

Notebook: `colab_notebooks/01_NACA0012_Airfoil.ipynb`. Solver: `code/exp01_airfoil_fvm_simple.py`.

2 Experiment 2: Fluid Flow and Heat Transfer in a Double-Pipe Heat Exchanger

Aim

To analyse conjugate heat transfer between hot and cold laminar streams separated by a conducting wall.

Governing Model

Fluid zones:

$$\rho c_p \, \mathbf{u} \cdot \nabla T = \nabla \cdot (k_f \nabla T). \tag{3}$$

Solid wall:

$$\nabla \cdot (k_s \nabla T) = 0. \tag{4}$$

At each interface, temperature and normal heat flux are continuous.

Boundary Conditions

Specified hot and cold inlet temperatures; zero-gradient outlet temperatures; adiabatic external boundaries; continuity of temperature and heat flux at fluid-solid interfaces.

Numerical Method

Steady finite-volume convection-diffusion in two fluid zones coupled to conduction in the intervening wall. Prescribed parabolic velocity profiles isolate the conjugate thermal problem.

> **Model scope.** Two-dimensional stacked-domain analogue with prescribed velocities; not a full three-dimensional exchanger simulation.

Run in Google Colab

Upload the notebook below to Google Colab, save a copy in Drive, select the run mode, and use **Runtime > Run all**.

```
colab_notebooks/02_Double_Pipe_CHT.ipynb
```

Argument order

```
nx ny_cold ny_wall ny_hot Re_hot Re_cold ks_by_kf counter_flow max_iterations Pr T_hot_in
    T_cold_in U_hot U_cold
```

Primary outputs

```
double_pipe_cht.vtk, history.csv, summary.txt
```

Procedure

1. Run the notebook first in `quick` mode to verify the Colab workflow.
2. Change to `lab` mode and select **Runtime > Run all**.
3. Confirm that the solver completed without exceptions and read the summary file.
4. Inspect the residual, error, conservation, or force history plotted in the notebook.
5. Use the inline VTK field preview to produce the required visualization. Downloaded VTK files may be opened in ParaView as an optional extension.
6. Modify only the parameter cell for the compulsory assignment; retain the original base-case results.

Observations to Record

Outlet temperatures, heat-transfer rate, effectiveness, residual, relative energy imbalance, and temperature contours.

Compulsory Assignment

Run Reynolds numbers 50, 100, and 200. Repeat one case for wall-to-fluid conductivity ratios 1 and 100. Compare outlet temperatures, heat transfer, effectiveness, and energy balance.

Implementation Files

Notebook: `colab_notebooks/02_Double_Pipe_CHT.ipynb`. Solver: `code/exp02_double_pipe_cht_fvm.py`.

3 Experiment 3: Internal Flow and Heat Transfer in a Centrifugal Pump

Aim

To study centrifugal pressure rise, absolute and relative velocities, pump head, and temperature change in a simplified rotating annular passage.

Governing Model

The leading-order centrifugal pressure rise is

$$\Delta p \approx \frac{1}{2}\rho\Omega^2(r_2^2 - r_1^2). \tag{5}$$

Thermal transport is represented by

$$\rho c_p \, \mathbf{u}_{rel} \cdot \nabla T = k\nabla^2 T + S_T. \tag{6}$$

Boundary Conditions

Specified inlet temperature and radial inflow at the eye; outward radial transport; periodic azimuthal direction; wall heat exchange represented through a distributed source.

Numerical Method

Polar single-rotating-frame reduced-order model with prescribed radial and tangential velocity components, centrifugal pressure rise, wall heat transfer, and loss heating.

Model scope. Reduced-order annular model; not blade-resolved pump CFD and not suitable for efficiency-map prediction.

Run in Google Colab

Upload the notebook below to Google Colab, save a copy in Drive, select the run mode, and use **Runtime > Run all**.

```
colab_notebooks/03_Centrifugal_Pump_SRF.ipynb
```

Argument order

```
nr ntheta rpm Re T_in T_wall iterations swirl wall_HTC loss_fraction
```

Primary outputs

```
pump_annulus_srf.vtk, pump_boundaries.vtk, pump_flow_path_guides.vtk, pump_blade_guides.vtk,
    pump_stream_seeds.vtk, pump_history.csv, pump_summary.txt
```

Procedure

1. Run the notebook first in `quick` mode to verify the Colab workflow.
2. Change to `lab` mode and select **Runtime > Run all**.
3. Confirm that the solver completed without exceptions and read the summary file.
4. Inspect the residual, error, conservation, or force history plotted in the notebook.

5. Use the inline VTK field preview to produce the required visualization. Downloaded VTK files may be opened in ParaView as an optional extension.
6. Modify only the parameter cell for the compulsory assignment; retain the original base-case results.

Observations to Record

RPM, pressure rise, pump head, inlet and outlet temperatures, maximum absolute speed, relative velocity, and thermal residual.

Compulsory Assignment

Run 1000, 2000, and 3000 rpm at fixed Reynolds number. Plot pump head against RPM squared and assess the expected scaling.

Implementation Files

Notebook: `colab_notebooks/03_Centrifugal_Pump_SRF.ipynb`. Solver: `code/exp03_centrifugal_pump_srf.py`.

4 Experiment 4: Karman Vortex Shedding over a Circular Cylinder

Aim

To resolve alternating vortex shedding behind a circular cylinder and estimate the Strouhal number.

Governing Model

The D2Q9 BGK lattice Boltzmann equation is

$$f_i(\mathbf{x} + \mathbf{c}_i \Delta t, t + \Delta t) = f_i(\mathbf{x}, t) - \frac{1}{\tau}\left(f_i - f_i^{eq}\right), \tag{7}$$

with lattice viscosity $\nu = c_s^2(\tau - 1/2)$. The shedding frequency is reported as

$$St = \frac{fD}{U_\infty}. \tag{8}$$

Boundary Conditions

Prescribed low-Mach inlet velocity; open outlet treatment; wall treatment on upper/lower boundaries; no-slip bounce-back at the cylinder.

Numerical Method

D2Q9 BGK lattice Boltzmann method with bounce-back cylinder treatment, momentum-exchange drag/lift evaluation, and Fourier estimation of shedding frequency.

> **Model scope.** Weakly compressible lattice-unit model. The supplied laboratory case is validated for periodic shedding near $St \approx 0.167$ at $Re = 100$.

Run in Google Colab

Upload the notebook below to Google Colab, save a copy in Drive, select the run mode, and use **Runtime > Run all**.

```
colab_notebooks/04_Cylinder_Vortex_Shedding.ipynb
```

Argument order

```
nx ny Re outer_steps numerical_iterations output_every
```

Primary outputs

```
cylinder_vortex_*.vtk, cylinder_vortex_final.vtk, cylinder_wall.vtk, cylinder_stream_seeds.vtk
    , cylinder_history.csv, cylinder_summary.txt
```

Procedure

1. Run the notebook first in `quick` mode to verify the Colab workflow.
2. Change to `lab` mode and select **Runtime > Run all**.
3. Confirm that the solver completed without exceptions and read the summary file.
4. Inspect the residual, error, conservation, or force history plotted in the notebook.
5. Use the inline VTK field preview to produce the required visualization. Downloaded VTK files may be opened in ParaView as an optional extension.
6. Modify only the parameter cell for the compulsory assignment; retain the original base-case results.

Observations to Record

Lift and drag histories, mean drag, lift amplitude, estimated Strouhal number, density deviation, vorticity field, and wake periodicity.

Compulsory Assignment

Use the laboratory case at $Re = 100$. Verify periodic shedding and compare the estimated St with the expected range 0.14–0.20. Show one full shedding cycle.

Implementation Files

Notebook: `colab_notebooks/04_Cylinder_Vortex_Shedding.ipynb`. Solver: `code/exp04_cylinder_karman_vortex.py`.

5 Experiment 5: Transient Couette Flow: Explicit and Implicit Formulations

Aim

To compare explicit FTCS and implicit backward-Euler solutions of transient viscous momentum diffusion against the analytical series solution.

Governing Model

$$\frac{\partial u}{\partial t} = \nu \frac{\partial^2 u}{\partial y^2}, \qquad \nu \frac{\Delta t}{\Delta y^2} = Fo. \tag{9}$$

For FTCS, stability requires $Fo \leq 1/2$.

Boundary Conditions

Initially stationary fluid; lower plate fixed; upper plate impulsively set to speed U; no pressure gradient.

Numerical Method

One-dimensional finite-difference discretization. FTCS is used for the explicit solution and a tridiagonal backward-Euler system is used for the implicit solution.

Model scope. Canonical one-dimensional diffusion benchmark with an analytical reference.

Run in Google Colab

Upload the notebook below to Google Colab, save a copy in Drive, select the run mode, and use **Runtime > Run all**.

```
colab_notebooks/05_Transient_Couette_Flow.ipynb
```

Argument order

```
ny H U nu t_end Fo_explicit Fo_implicit screen_every
```

Primary outputs

```
couette_field.vtk, couette_plates.vtk, couette_profiles.csv, couette_history.csv,
    couette_summary.txt
```

Procedure

1. Run the notebook first in `quick` mode to verify the Colab workflow.
2. Change to `lab` mode and select **Runtime > Run all**.
3. Confirm that the solver completed without exceptions and read the summary file.
4. Inspect the residual, error, conservation, or force history plotted in the notebook.
5. Use the inline VTK field preview to produce the required visualization. Downloaded VTK files may be opened in ParaView as an optional extension.
6. Modify only the parameter cell for the compulsory assignment; retain the original base-case results.

Observations to Record

Velocity profiles, explicit and implicit time steps, relative L2 errors, maximum errors, stability, and computational effort.

Compulsory Assignment

Run explicit Fourier numbers 0.25, 0.45, and 0.55. Explain the stability boundary and compare both schemes at the same physical time.

Implementation Files

Notebook: `colab_notebooks/05_Transient_Couette_Flow.ipynb`. Solver: `code/exp05_couette_explicit_implicit.py`.

6 Experiment 6: Two-Dimensional Plate Heat Conduction

Aim

To compare Jacobi and line-TDMA/SOR solutions of the steady two-dimensional Laplace equation against an analytical solution.

Governing Model

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0. \tag{10}$$

The analytical reference is

$$T = T_L \frac{\sinh[\pi(L-x)/H]}{\sinh(\pi L/H)} \sin\left(\frac{\pi y}{H}\right). \tag{11}$$

Boundary Conditions

$T(0, y) = T_L \sin(\pi y/H)$; the right, top, and bottom boundaries are held at zero temperature.

Numerical Method

Second-order finite differences with point Jacobi iteration and line-implicit TDMA/SOR iteration.

Model scope. Steady Laplace benchmark with an exact analytical reference.

Run in Google Colab

Upload the notebook below to Google Colab, save a copy in Drive, select the run mode, and use **Runtime > Run all**.

```
colab_notebooks/06_Plate_Conduction_2D.ipynb
```

Argument order

```
nx ny L H T_left max_iterations tolerance omega screen_every
```

Primary outputs

```
plate_conduction_2d.vtk, plate_boundaries.vtk, plate_conduction_profiles.csv,
    plate_conduction_history.csv, plate_conduction_summary.txt
```

Procedure

1. Run the notebook first in `quick` mode to verify the Colab workflow.
2. Change to `lab` mode and select **Runtime > Run all**.
3. Confirm that the solver completed without exceptions and read the summary file.
4. Inspect the residual, error, conservation, or force history plotted in the notebook.
5. Use the inline VTK field preview to produce the required visualization. Downloaded VTK files may be opened in ParaView as an optional extension.
6. Modify only the parameter cell for the compulsory assignment; retain the original base-case results.

Observations to Record

Iteration counts, final residuals, relative L2 errors, centreline profiles, and influence of relaxation factor and grid size.

Compulsory Assignment

Compare both methods on at least two grids. Report convergence rate, analytical error, and the effect of SOR relaxation factor.

Implementation Files

Notebook: `colab_notebooks/06_Plate_Conduction_2D.ipynb`. Solver: `code/exp06_plate_conduction_explicit_implicit.py`.

7 Experiment 7: One-Dimensional Wave Propagation in a Still Lake

Aim

To compare numerical diffusion, dispersion, stability, and conservation for first-order upwind and Lax-Wendroff schemes.

Governing Model

For small perturbations of depth H,

$$\frac{\partial \eta}{\partial t} + H\frac{\partial u}{\partial x} = 0, \tag{12}$$

$$\frac{\partial u}{\partial t} + g\frac{\partial \eta}{\partial x} = 0, \tag{13}$$

with wave speed $c = \sqrt{gH}$ and $CFL = c\Delta t/\Delta x$.

Boundary Conditions

Periodic one-dimensional domain with an initially localized Gaussian surface perturbation.

Numerical Method

Conservative first-order upwind and second-order Lax-Wendroff time marching on the linearized shallow-water system.

Model scope. Linearized small-amplitude shallow-water benchmark.

Run in Google Colab

Upload the notebook below to Google Colab, save a copy in Drive, select the run mode, and use **Runtime > Run all**.

```
colab_notebooks/07_Shallow_Water_Wave.ipynb
```

Argument order

```
nx L H amplitude sigma x0 t_end CFL output_every gravity vertical_scale nz_visual
```

Primary outputs

```
OPEN_THIS_lake_visual_final.vtk, OPEN_THIS_lake_surface_ribbon_final.vtk, lake_space_time.vtk,
    lake_wave_profiles.csv, lake_wave_history.csv, lake_wave_summary.txt
```

Procedure

1. Run the notebook first in `quick` mode to verify the Colab workflow.
2. Change to `lab` mode and select **Runtime > Run all**.
3. Confirm that the solver completed without exceptions and read the summary file.
4. Inspect the residual, error, conservation, or force history plotted in the notebook.
5. Use the inline VTK field preview to produce the required visualization. Downloaded VTK files may be opened in ParaView as an optional extension.
6. Modify only the parameter cell for the compulsory assignment; retain the original base-case results.

Observations to Record

Wave amplitude, phase error, relative L2 error, mass conservation, numerical diffusion, oscillations, and CFL dependence.

Compulsory Assignment

Run CFL values 0.4, 0.8, and 1.1. Compare stability, amplitude loss, phase error, and conservation for both schemes.

Implementation Files

Notebook: `colab_notebooks/07_Shallow_Water_Wave.ipynb`. Solver: `code/exp07_lake_wave_hyperbolic.py`.

8 Experiment 8: Rayleigh-Taylor Instability using a VOF-Style Method

Aim

To study the growth of a perturbed heavy-over-light interface while monitoring phase-fraction boundedness and conservation.

Governing Model

The transported phase indicator α satisfies

$$\frac{\partial \alpha}{\partial t} + \nabla \cdot (\alpha \mathbf{u}) = 0, \tag{14}$$

while the Boussinesq momentum equation contains a buoyancy term proportional to the Atwood number

$$A = \frac{\rho_h - \rho_l}{\rho_h + \rho_l}. \tag{15}$$

Boundary Conditions

No-slip impermeable box boundaries; initially perturbed interface with heavier fluid above lighter fluid.

Numerical Method

Projection flow solver coupled to conservative upwind phase-fraction transport, clipping, and conservative redistribution to maintain $0 \leq \alpha \leq 1$.

Model scope. Educational bounded VOF-style model; not a geometric reconstruction production VOF solver.

Run in Google Colab

Upload the notebook below to Google Colab, save a copy in Drive, select the run mode, and use **Runtime > Run all**.

```
colab_notebooks/08_Rayleigh_Taylor_VOF.ipynb
```

Argument order

```
nx ny Atwood gravity nu t_end max_steps poisson_iterations output_every
```

Primary outputs

```
rti_*.vtk, rti_final.vtk, rti_interface_final.vtk, rti_domain_boundary.vtk, rti_history.csv,
    rti_summary.txt
```

Procedure

1. Run the notebook first in `quick` mode to verify the Colab workflow.
2. Change to `lab` mode and select **Runtime > Run all**.
3. Confirm that the solver completed without exceptions and read the summary file.
4. Inspect the residual, error, conservation, or force history plotted in the notebook.
5. Use the inline VTK field preview to produce the required visualization. Downloaded VTK files may be opened in ParaView as an optional extension.
6. Modify only the parameter cell for the compulsory assignment; retain the original base-case results.

Observations to Record

Spike and bubble positions, mixing width, divergence, mass error, minimum and maximum phase fraction, and interface shape.

Compulsory Assignment

Run Atwood numbers 0.2, 0.5, and 0.8. Compare spike, bubble, mixing width, boundedness, and conservation.

Implementation Files

Notebook: `colab_notebooks/08_Rayleigh_Taylor_VOF.ipynb`. Solver: `code/exp08_rayleigh_taylor_vof.py`.

9 Experiment 9: Lid-Driven Cavity using Vorticity-Streamfunction

Aim

To solve the incompressible cavity problem and analyse the primary recirculation using vorticity and streamfunction.

Governing Model

$$\nabla^2\psi = -\omega, \tag{16}$$

$$u = \frac{\partial\psi}{\partial y}, \qquad v = -\frac{\partial\psi}{\partial x}, \tag{17}$$

$$\frac{\partial\omega}{\partial t} + u\frac{\partial\omega}{\partial x} + v\frac{\partial\omega}{\partial y} = \frac{1}{Re}\nabla^2\omega. \tag{18}$$

Boundary Conditions

Unit square; stationary side and bottom walls; top wall moving at prescribed lid speed; zero streamfunction on all walls.

Numerical Method

Explicit upwind vorticity transport, red-black SOR solution of the streamfunction Poisson equation, and Thom-type wall-vorticity conditions.

> **Model scope.** The student must perform grid and benchmark comparison; the pack does not claim automatic grid independence.

Run in Google Colab

Upload the notebook below to Google Colab, save a copy in Drive, select the run mode, and use **Runtime > Run all**.

```
colab_notebooks/09_Lid_Driven_Cavity.ipynb
```

Argument order

```
nx ny Re U_lid max_steps output_every poisson_iterations tolerance CFL SOR
```

Primary outputs

```
cavity_*.vtk, cavity_final.vtk, cavity_boundaries.vtk, cavity_stream_seeds.vtk,
    cavity_centerline_profiles.csv, cavity_history.csv, cavity_summary.txt
```

Procedure

1. Run the notebook first in `quick` mode to verify the Colab workflow.
2. Change to `lab` mode and select **Runtime > Run all**.
3. Confirm that the solver completed without exceptions and read the summary file.
4. Inspect the residual, error, conservation, or force history plotted in the notebook.
5. Use the inline VTK field preview to produce the required visualization. Downloaded VTK files may be opened in ParaView as an optional extension.
6. Modify only the parameter cell for the compulsory assignment; retain the original base-case results.

Observations to Record

Primary vortex centre, minimum streamfunction, centreline velocity profiles, vorticity-change norm, Poisson residual, and grid dependence.

Compulsory Assignment

Run Reynolds numbers 100 and 400. Compare centreline velocities, primary-vortex position, convergence, and secondary-corner-vortex development.

Implementation Files

Notebook: `colab_notebooks/09_Lid_Driven_Cavity.ipynb`. Solver: `code/exp09_lid_cavity_psi_omega.py`.

10 Experiment 10: Turbulent Pipe-Flow Model Comparison

Aim

To compare radial eddy-viscosity closures using mean velocity, friction factor, pressure drop, and turbulent viscosity.

Governing Model

For fully developed axisymmetric mean flow,

$$\frac{1}{r}\frac{d}{dr}\left[r(\nu+\nu_t)\frac{dU}{dr}\right]=\frac{1}{\rho}\frac{dp}{dx}. \tag{19}$$

The closures differ through the model for ν_t.

Boundary Conditions

Axis symmetry at the centreline, no-slip at the wall, and pressure gradient adjusted to obtain unit bulk velocity.

Numerical Method

Radial finite-volume momentum solution for laminar, mixing-length, Van-Driest-damped, equilibrium k-ε, and equilibrium k-ω/SST surrogate closures.

> **Model scope.** Equilibrium radial closures; not complete two-equation RANS transport simulations.

Run in Google Colab

Upload the notebook below to Google Colab, save a copy in Drive, select the run mode, and use **Runtime > Run all**.

```
colab_notebooks/10_Turbulent_Pipe_Models.ipynb
```

Argument order

```
n_radial vtk_nx vtk_ny Re_min Re_max n_Re max_iterations tolerance
```

Primary outputs

```
pipe2d_compare_Re*.vtk, pipe2d_profiles_Re*.csv, pipe2d_rans_summary.csv, pipe2d_rans_summary.
    txt, pipe2d_walls.vtk
```

Procedure

1. Run the notebook first in `quick` mode to verify the Colab workflow.
2. Change to `lab` mode and select **Runtime > Run all**.
3. Confirm that the solver completed without exceptions and read the summary file.
4. Inspect the residual, error, conservation, or force history plotted in the notebook.
5. Use the inline VTK field preview to produce the required visualization. Downloaded VTK files may be opened in ParaView as an optional extension.
6. Modify only the parameter cell for the compulsory assignment; retain the original base-case results.

Observations to Record

Velocity profile fullness, friction factor, Blasius and Haaland references, wall shear, pressure drop, and maximum turbulent-viscosity ratio.

Compulsory Assignment

Compare all model profiles at $Re = 10^4$, approximately 3.16×10^4, and 10^5. Explain which closure gives the fullest profile and compare friction factors with reference correlations.

Implementation Files

Notebook: `colab_notebooks/10_Turbulent_Pipe_Models.ipynb`. Solver: `code/exp10_pipe_turbulence_models.py`.

11 Experiment 11: Generic Convection-Diffusion Transport

Aim

To compare scalar transport in forced flat-plate, forced channel, and natural-convection surrogate velocity fields.

Governing Model

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\mathbf{u}\phi) = \nabla \cdot (\Gamma \nabla \phi) + S_\phi, \tag{20}$$

with cell Peclet number $Pe = |\mathbf{u}|\Delta x/\Gamma$.

Boundary Conditions

Case-dependent thermal boundaries: heated flat plate, heated channel walls, or hot/cold vertical walls for the recirculating surrogate.

Numerical Method

Finite-volume-style transient iteration with upwind convection and central diffusion for three prescribed velocity fields.

> **Model scope.** The natural-convection case uses a prescribed recirculating field; it is not a coupled Boussinesq solver.

Run in Google Colab

Upload the notebook below to Google Colab, save a copy in Drive, select the run mode, and use **Runtime > Run all**.

```
colab_notebooks/11_Convection_Diffusion.ipynb
```

Argument order

```
mode nx ny Pe iterations output_every; mode = all, flat, pipe, or natural
```

Primary outputs

```
forced_flat_plate.vtk, forced_pipe_channel.vtk, natural_vertical_plate_surrogate.vtk, case
    history/profile CSV files, convective_transport_summary.csv, convective_transport_summary.
    txt
```

Procedure

1. Run the notebook first in `quick` mode to verify the Colab workflow.
2. Change to `lab` mode and select **Runtime > Run all**.
3. Confirm that the solver completed without exceptions and read the summary file.
4. Inspect the residual, error, conservation, or force history plotted in the notebook.
5. Use the inline VTK field preview to produce the required visualization. Downloaded VTK files may be opened in ParaView as an optional extension.
6. Modify only the parameter cell for the compulsory assignment; retain the original base-case results.

Observations to Record

Residual, outlet or mean temperature, average wall flux, scalar boundary-layer thickness, local Peclet number, and field shape.

Compulsory Assignment

Run all three cases at Peclet numbers 50, 200, and 1000. Compare field shape, boundary-layer thickness, wall flux, outlet or mean temperature, and convergence.

Implementation Files

Notebook: `colab_notebooks/11_Convection_Diffusion.ipynb`. Solver: `code/exp11_convection_diffusion.py`.

A Google Colab Workflow and Troubleshooting

Problem	Corrective action
Notebook was not saved	Select **File** > **Save a copy in Drive** before changing parameters.
Runtime disconnected	Reconnect and select **Runtime** > **Run all**. Temporary files from the previous runtime are not recoverable.
A result ZIP was not downloaded	Re-run the final download cell with `DOWNLOAD_RESULTS = True`.
Notebook run is slow	Use `quick` only for checking. Use the prescribed `lab` grid once for final results. Free Colab CPU performance varies.
Old results appear	Re-run the solver cell. It deletes its own previous output folder before execution.
Inline contour is missing	Confirm that the solver completed and generated the stated VTK file. Then re-run the field-preview cell.
Residual does not fall	Check the custom parameter values, grid size, stability controls, and whether a deliberately unstable case was selected.
Browser tab closed	Reopen the notebook from Google Drive. Re-run all cells because Colab runtime storage is temporary.

B Optional ParaView Extension

The basic laboratory workflow does not require ParaView. Each results ZIP nevertheless contains legacy VTK files. For optional advanced post-processing, open the final VTK file or transient file series in ParaView and use Contour, Glyph, Stream Tracer, or Plot Over Line as appropriate.

C Digital Package Contents

- `colab_notebooks/`: Start notebook and eleven self-contained experiment notebooks.
- `manual/`: Google Colab laboratory manual in PDF and editable LaTeX formats.
- `instructor_source/code/`: Auditable source of the eleven tested Python/NumPy solvers.
- `README_FIRST.md`: Student upload, execution, and download workflow.
- `COLAB_VALIDATION_REPORT.md`: Notebook structure and quick-execution verification.